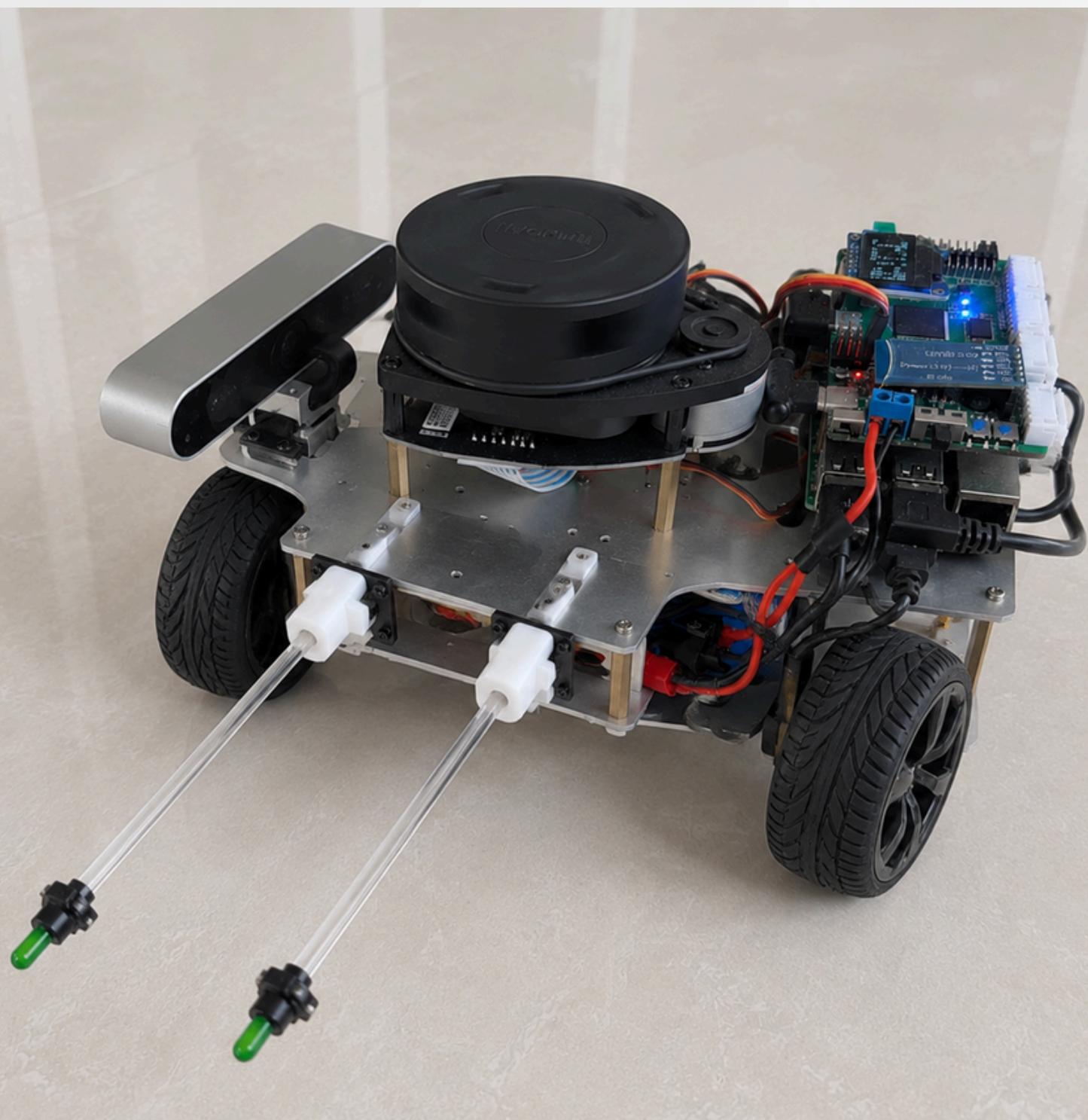


CANCELLATION OF SELF-SENSING DUE TO EGO-MOTION OF VIBRATION BASED MECHANOSENSORS



EE405
UNDERGRADUATE PROJECT I
GROUP : G31

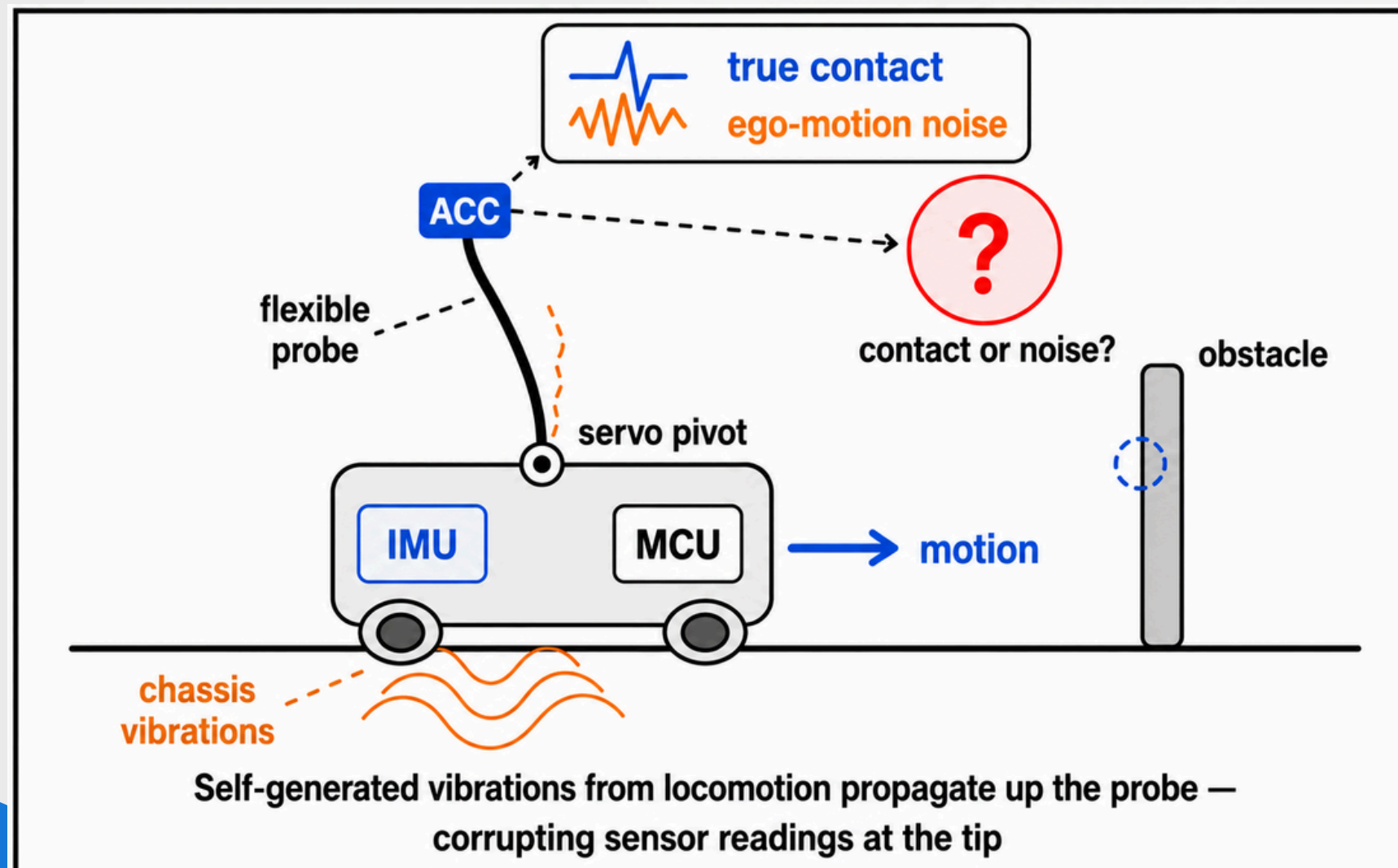
PROBLEM STATEMENT

Why Tactile Sensing?

- Crucial for autonomous navigation in low-light, occluded, or visually difficult environments.
- Cameras and LiDAR fail where flexible probe-based mechanosensors still function.

The Ego-Motion Problem

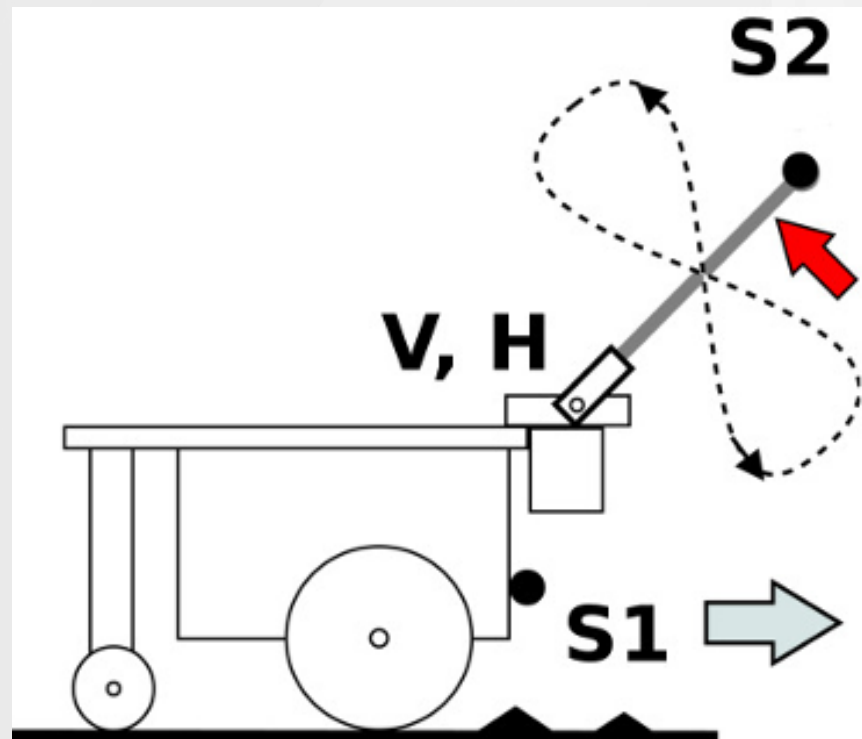
- Robot locomotion generates vibrations that propagate through its own chassis.
- These self-generated vibrations cause false readings on tip-mounted mechanosensors.
- Main Challenge: separating true external contact from self-induced noise.



PROJECT OBJECTIVES & DELIVERABLES

Core Objective:

Develop and implement an embedded signal-processing and machine learning pipeline to isolate true touch events from self-sensing noise in real time.



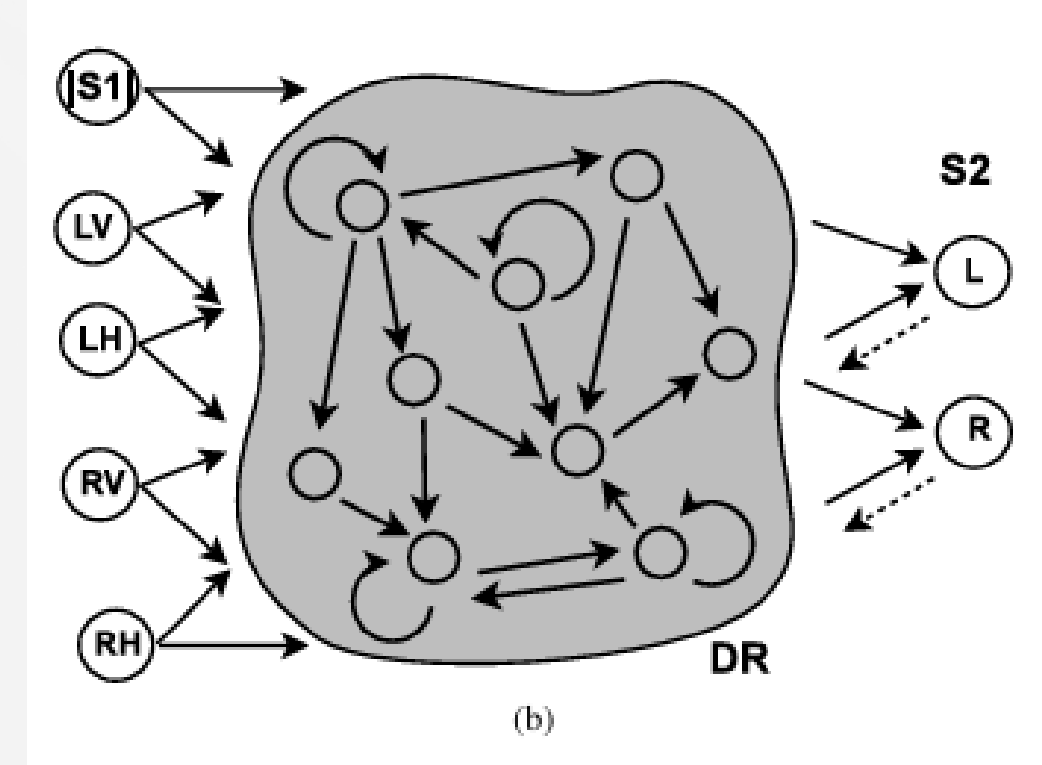
Bio-inspired Antenna

Construct a flexible-probe robotic antenna with a tip-mounted accelerometer.



Holonomic Platform

Integrate the sensor onto a mobile platform with an omnidirectional 4WD holonomic drive.



Embedded ESN Deployment

Deploy an Echo State Network forward model on a Raspberry Pi + Arduino Uno framework.

SCIENTIFIC FOUNDATIONS

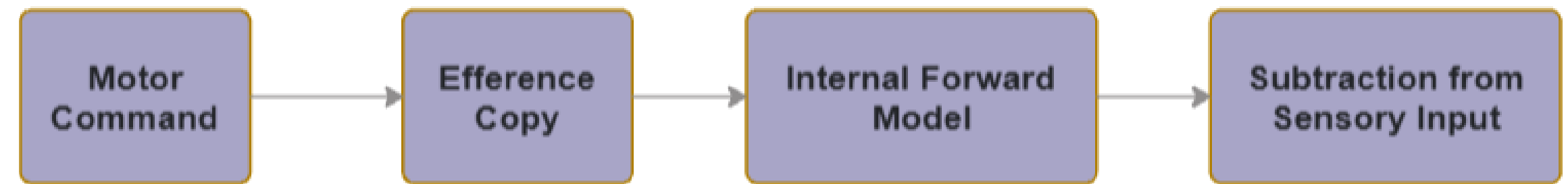
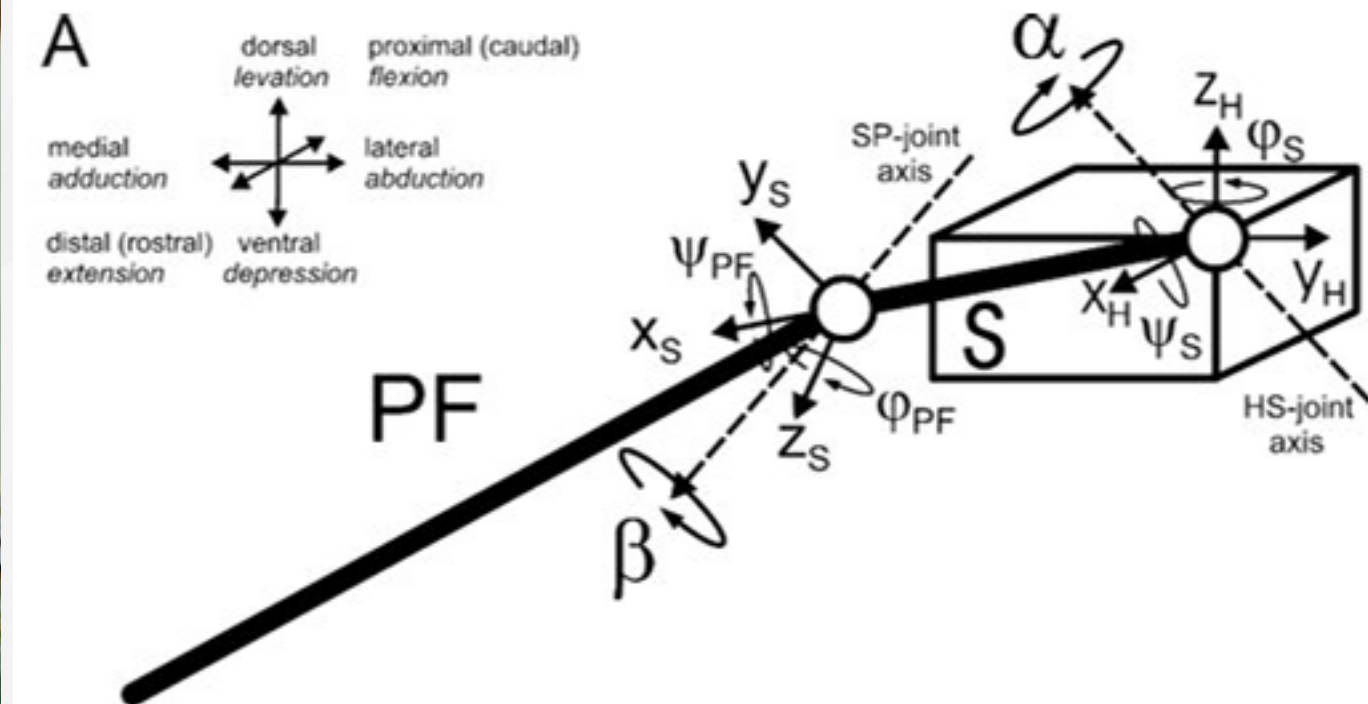
THE EFFERENCE COPY PRINCIPLE

Biological Inspiration:

Insects (crickets, cockroaches) use active antennal movement for spatial mapping during locomotion.

Phase-Dependent Sensitivity:

Antennal sensitivity varies dynamically with its movement phase cycle.



Replicated directly in our project by feeding motor telemetry into a predictive algorithm.

Predictive subtraction: the sensory center forecasts and removes expected self-generated motion, leaving only external touch signals.

CURRENT CHALLENGES IN ACTIVE SENSING

Research Gap

Challenge 1: Computing Limitations

Advanced algorithms are often too heavy to run in real-time on small mobile systems.

Challenge 2: Simplistic Testing

Prior systems are usually tested only in straight lines on perfectly flat surfaces.

Challenge 3: Sensor Input Noise Floor

Raw accelerometer data suffers from severe high-frequency background jitter, which can mask subtle touch details.

Our Approach

Deploying a lightweight Echo State Network that processes complex, multi-axis motion data instantly on the robot chassis.

Implementing a filter to clean the incoming signal stream before it reaches the ESN.



LITERARY REVIEW: COMPARATIVE ANALYSIS

Comparison of Noise Canceling Filters

Technique	Advantages	Limitations
Complementary Filter	Simple, Low computation	Lower accuracy
Kalman Filter	High accuracy, excellent noise reduction	Computationally expensive
Madgwick Filter	Lightweight, fast, Better for MPU6050	Primarily estimates orientation

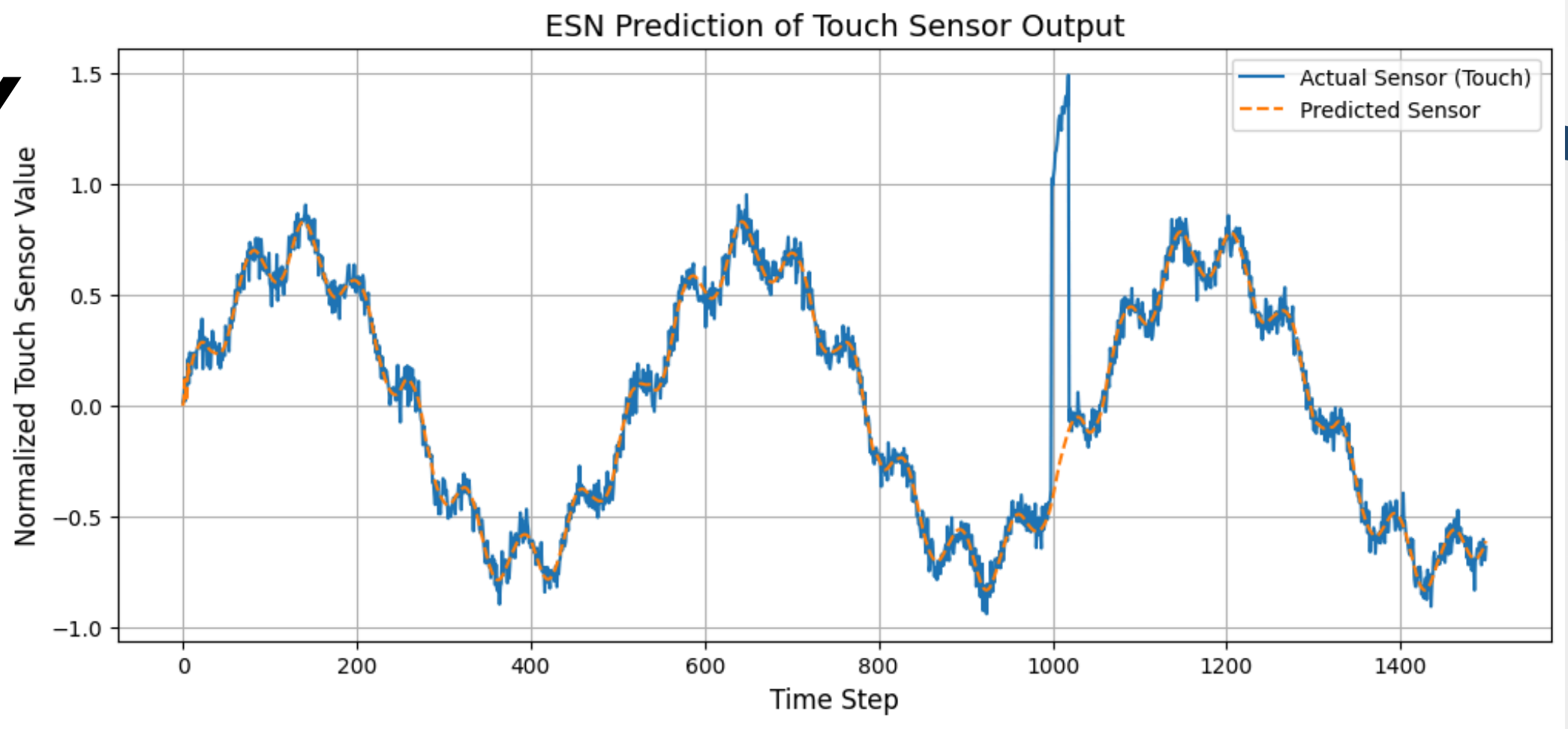
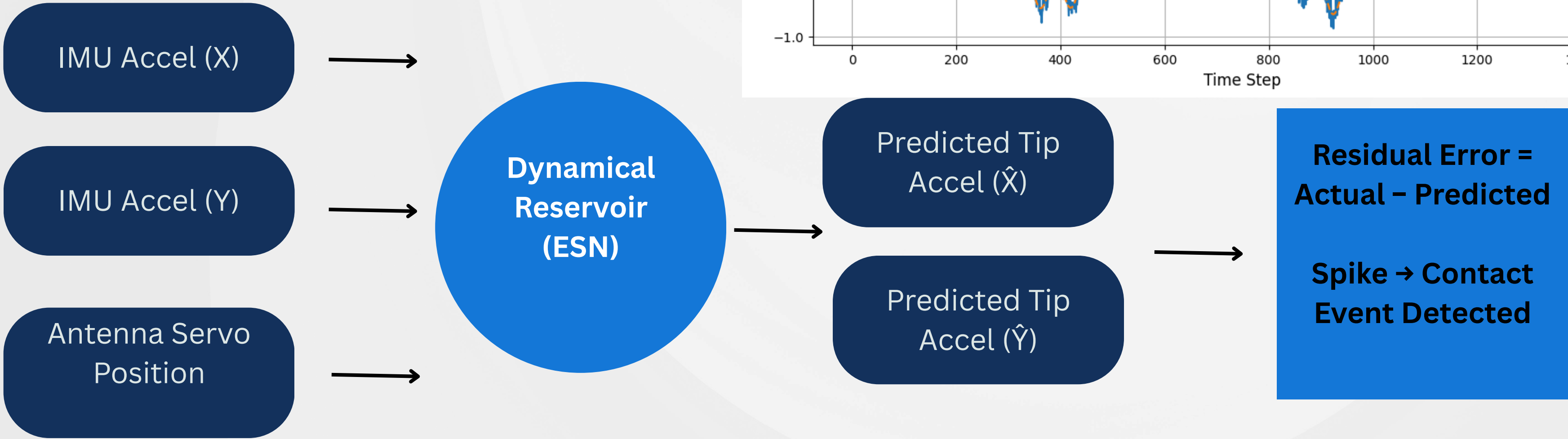
Machine Learning Model Comparison

Technique	Advantages	Limitation
ARMAX	Simple, interpretable, lightweight	Limited to linear dynamics
Echo State Network	Learns nonlinear temporal patterns, fast training	Reservoir tuning required
Deep Learning (CNN/LSTM)	Highest accuracy	Large dataset and high computational cost



PROPOSED METHODOLOGY & ARCHITECTURE

ESN Forward Model Architecture



Event Classification: a sudden residual deviation spike indicates a true external contact event, distinct from self-induced ego-motion noise.

IMPLEMENTATION FRAMEWORK & COMPONENT SELECTION

Sensing Nodes

Multi-axis IMU (MPU6050) for base reference, paired with a tip accelerometer on the flexible probe.

Raspberry Pi

High-level processing executes ESN matrix computations and real-time contact filtering.

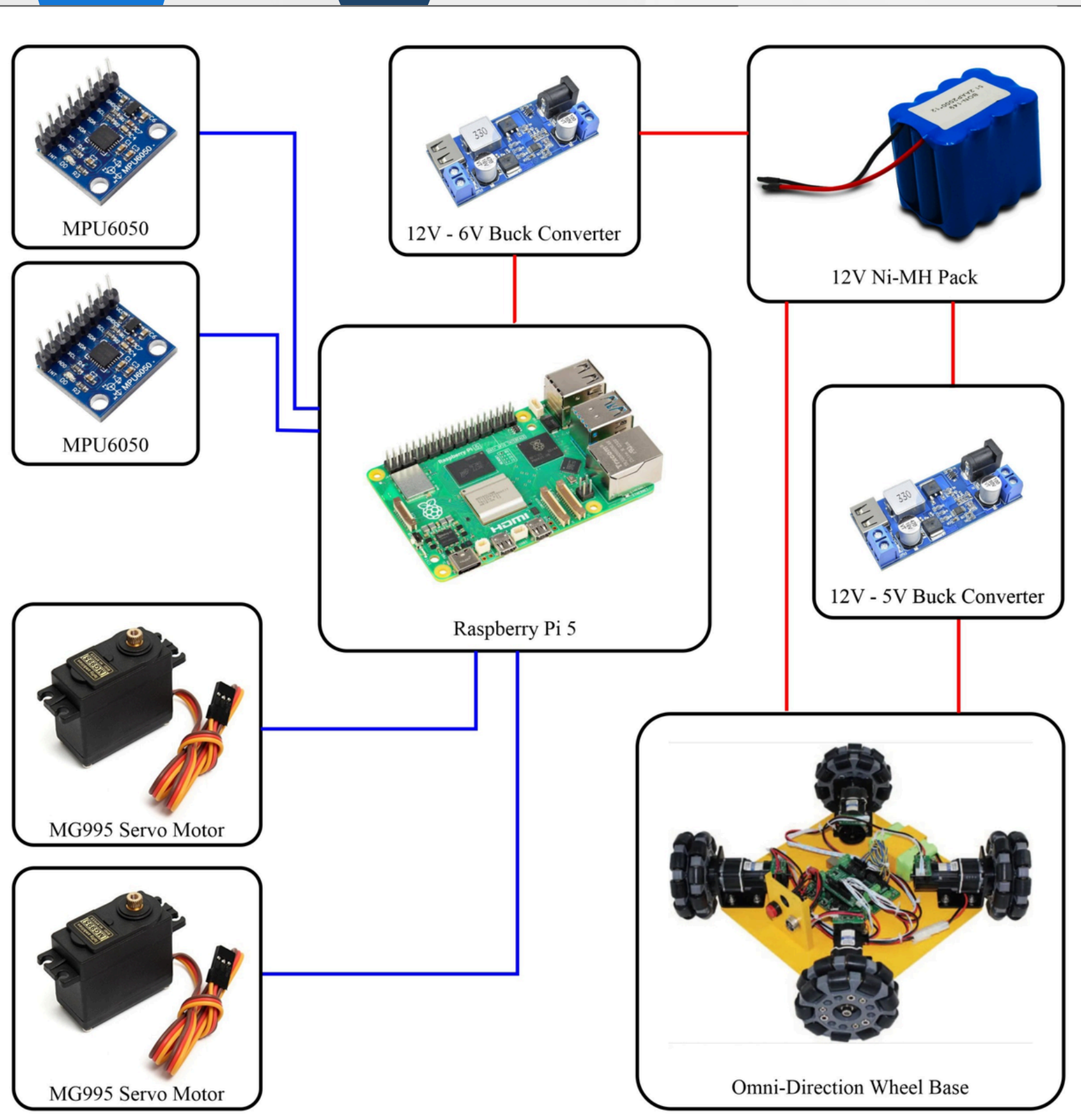
Omnidirectional Base

4WD holonomic wheel configuration driven by dedicated chassis motor controllers.

Arduino Uno

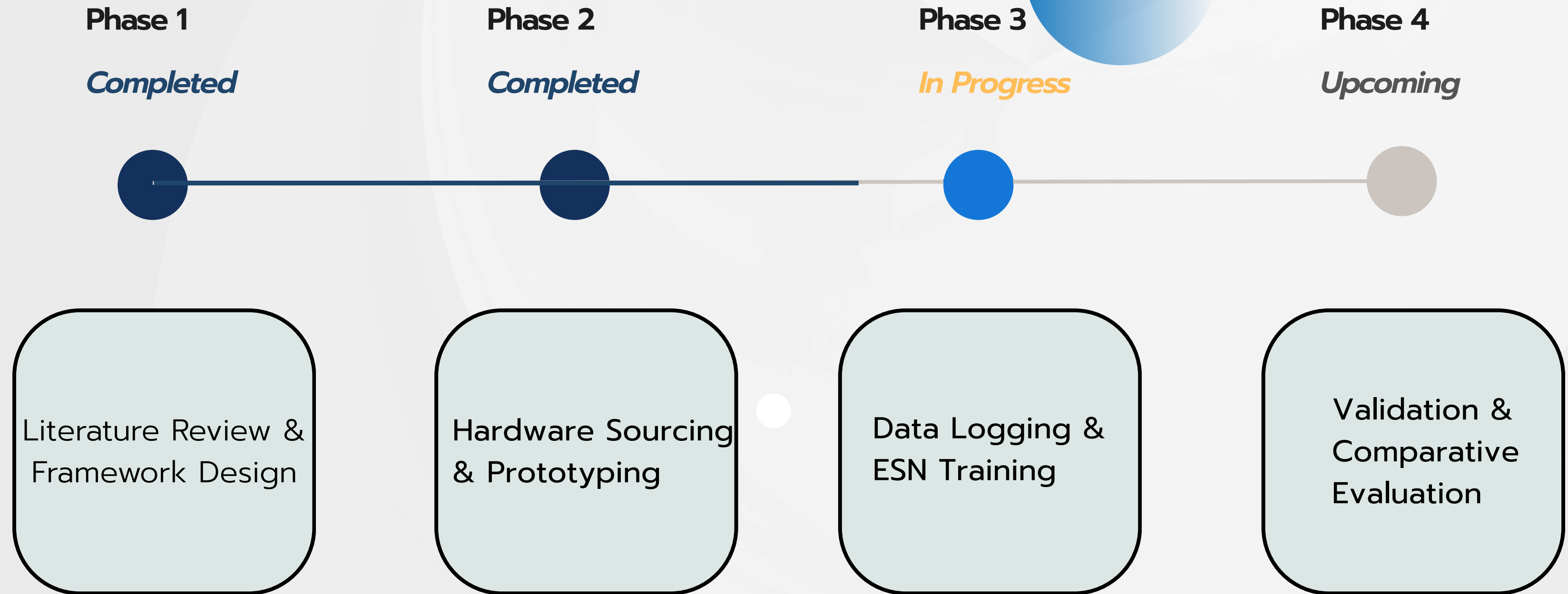
High-frequency, low-level data polling from sensors and servo telemetry tracking.

IMPLEMENTATION FRAMEWORK & COMPONENT SELECTION



- Blue: Signal/Data connections (I²C & PWM communication)
- Red: Power connections (12 V battery and regulated 6 V/5 V power supply)

PROJECT PLAN & MID-SEMESTER MILESTONES



Group is currently on schedule — Phase 3 (Data Logging & ESN Training) in progress.

Gantt Chart

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CURRENT PROGRESS & PRELIMINARY EVALUATION

Hardware Progress Status

Mechanical assembly and core data communication pipelines between Arduino are fully functional.

Algorithmic Preparation:

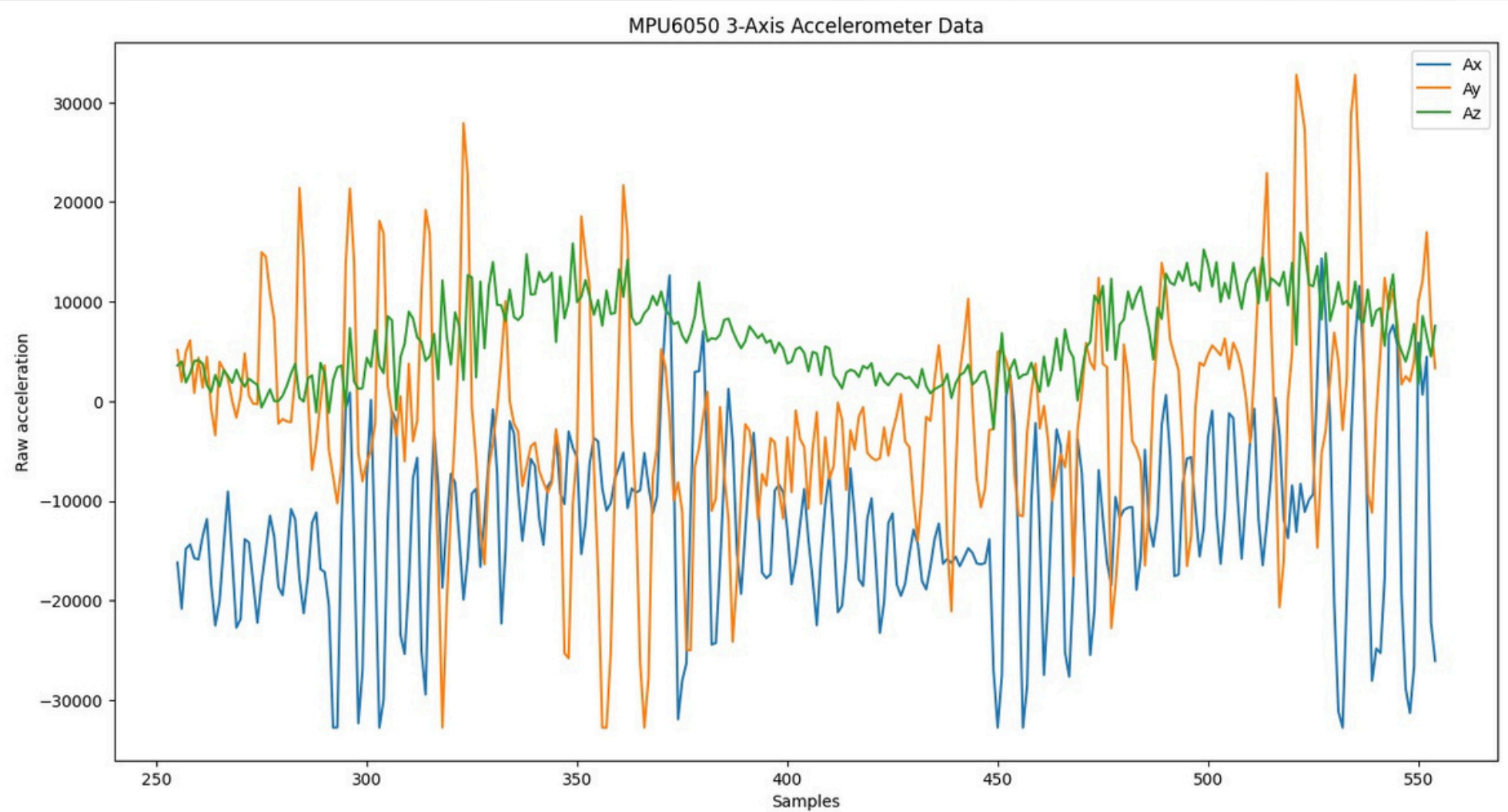
Reservoir structural equations and weight matrix sizes configured based on insect kinematic phase models.

Analytical Insights

Review of prior data confirms fixed-threshold linear methods drop below 70–80% noise rejection when sudden structural shifts or irregular surfaces are involved.

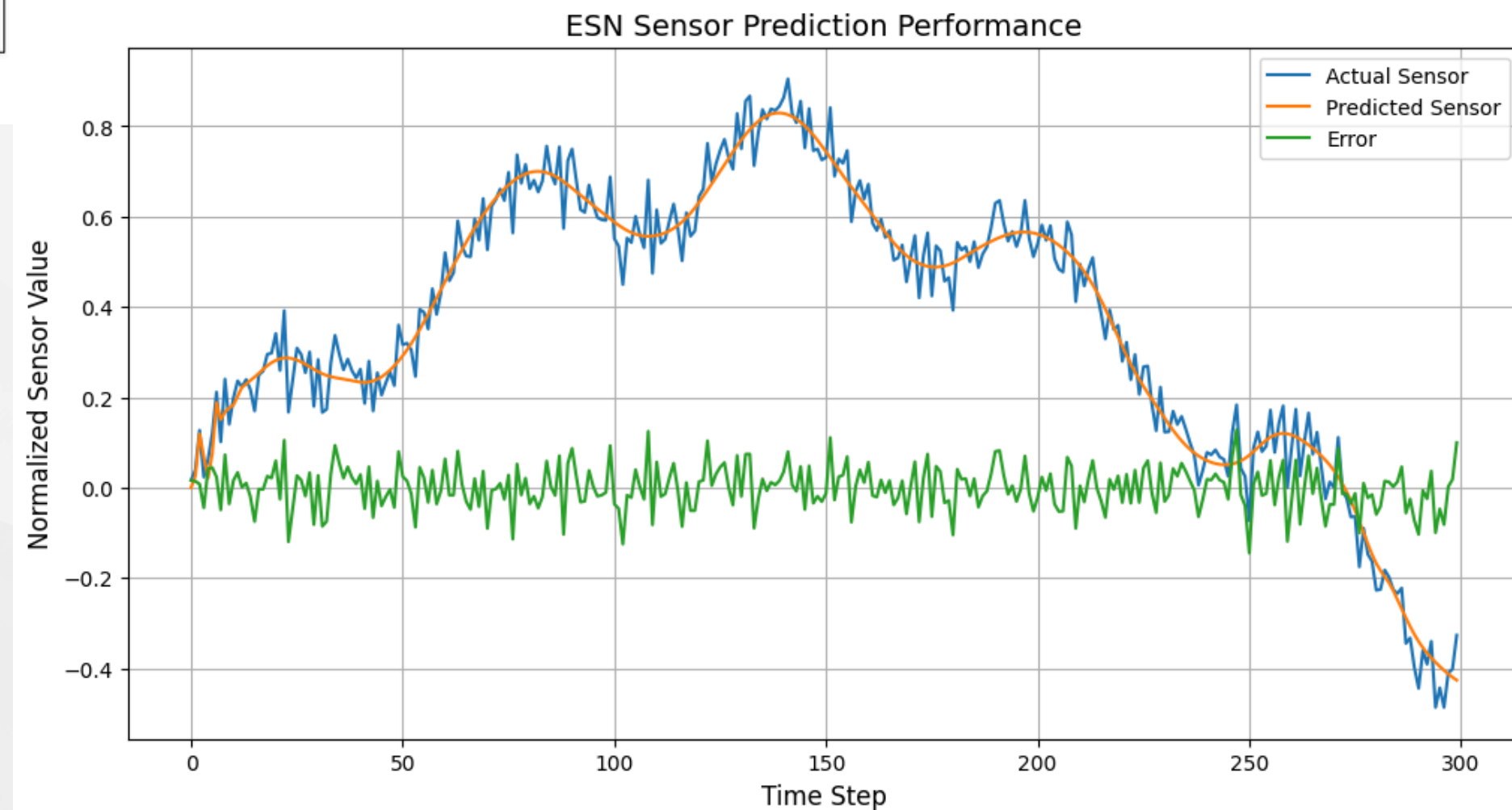
This reinforces the choice of an ESN model: reservoir computing is necessary to track non-linear, phase-dependent characteristics of a moving holonomic platform.

OBSERVATIONS/RESULTS



Results obtained from the 3-axes of the mounted Antenna before filtering.

Actual signal, Predicted Signal and the error signal of the sample ESN Model

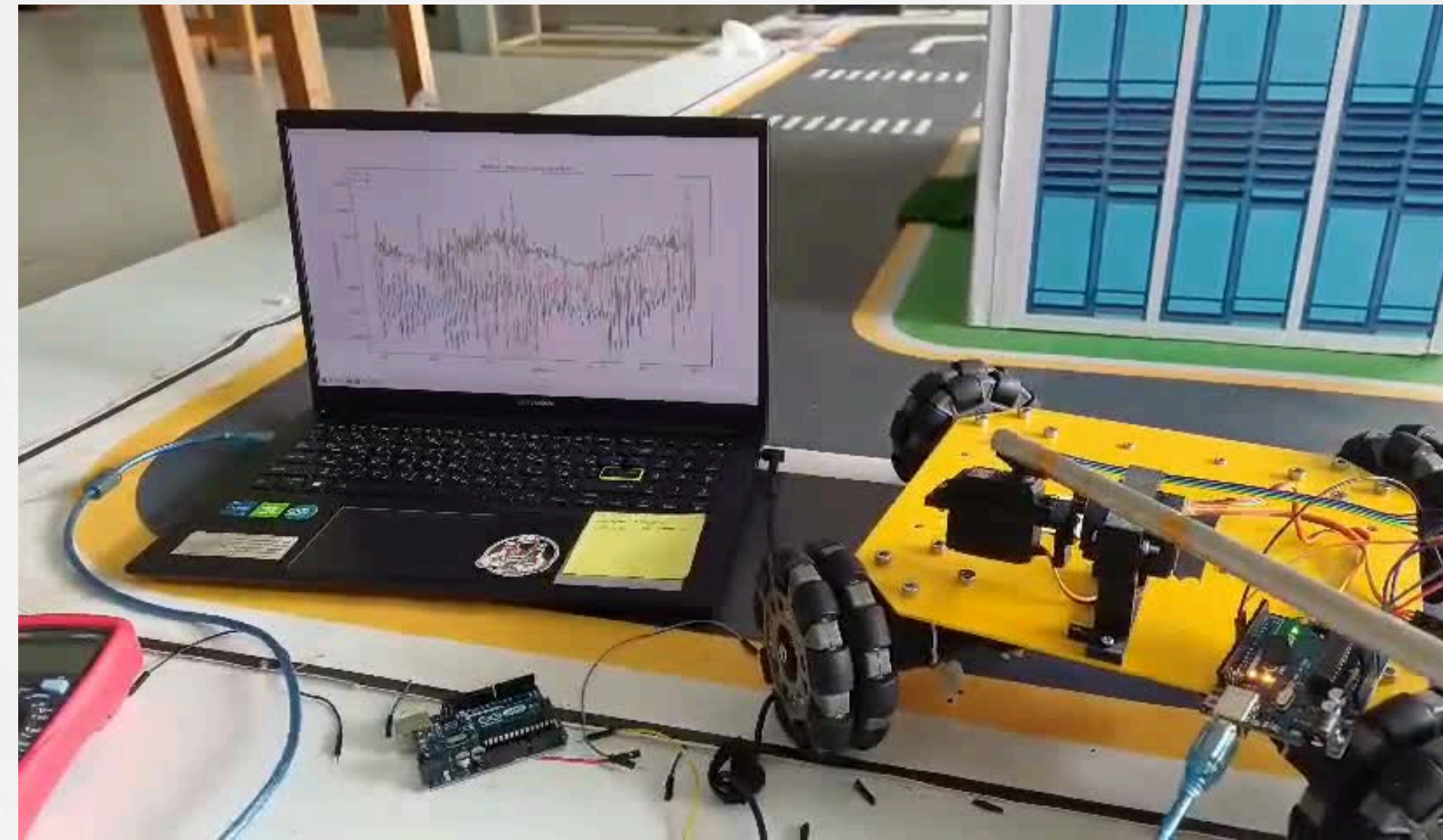


CONCLUSIONS & FUTURE MILESTONES

Preliminary Conclusion:

Echo State Networks offer the best balance of computational feasibility and non-linear noise cancellation for mobile embedded platforms.

- 1 Finalize the offline collection of self-sensing training data.
- 2 Train readout weights and flash the execution matrix onto the Raspberry Pi.
- 3 Run physical obstacle contact tests to map real-time sensitivity.



DELIVERABLES AND REFERENCES

Deliverables:

1. Literature Review:

https://drive.google.com/file/d/1Scmkz3GoWBiHw4eQhIUktYzD0q_mwWXN/view?usp=drive_link

2. Component Selection Report:

https://drive.google.com/file/d/1BPZq4nTgJhvT8DZAA7GpHTTjUrYqxoGK/view?usp=drive_link

1. Design Diagrams:

https://drive.google.com/file/d/1HqHS5aSH8MO8N4h5fhMfL4LpMrXgkYGe/view?usp=drive_link

2. Codes:

- Platform: https://drive.google.com/drive/folders/1JL1JS3Xzwc0cGVFVBLCdJ0xM6NcnPURm?usp=drive_link
- Accelerometer Data Acquisition: https://drive.google.com/drive/folders/1aTZOR2Bjfa_SXAcO-WrGmnUUu1FQrULW?usp=drive_link
- ESN: https://colab.research.google.com/drive/1guCraSH6m98zCZfai44IAiscpB1u0_Dr?usp=sharing

References:

- https://drive.google.com/drive/folders/1RYvGcrUYyHugNumTKyos3FHw_ixa7y9H?usp=drive_link

THANK YOU

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